Statistical protocol: matched comparisons, intervals, and bounded claims I

No “robust beats naive” claim is written before a statistical test produces it (Algorithm Gate I). Where the active-inference community has typically reported belief-fusion outcomes as single illustrative runs, we treat every headline contrast as a paired hypothesis test with effect-size estimation, multiple-testing deflation, confidence intervals, and an observed-effect design-power calculation — the reporting discipline expected by robust statistics and applied inference (Huber and Ronchetti 2009; Efron and Tibshirani 1993; Koehler et al. 2009; Morris et al. 2019; Loy and Korobova 2021; Nakagawa and Cuthill 2007; Benjamini and Hochberg 1995; Wasserstein and Lazar 2016). The protocol lives in `statistics.py`, sits at the analysis tier above the locked FedGVI core, and emits every number the results sections report; nothing is hard-coded (ISC-30).

Paired comparison and standardized effect size I

The headline claim is *paired*: across matched scenarios — same seed, same contamination — does the robust aggregator raise consensus accuracy over the naive log-linear pool? For the expanded review grid, the inferential unit is the seed: trials are nested within a seed/cell and are averaged before seed-level contrasts. The primary robustness sweep also retains a trial-level paired diagnostic, but its matched trial replicates are nested within one fixed seeded world and are not independent worlds. The seed-level review-grid estimand is therefore the robust-minus-naive accuracy difference after the declared trial reduction, not a contrast of two independently drawn group means. Each configured robust method is retained in every review-grid rate panel and has its own seed-level contrast, interval, and test; no pooled-selected curve or pooled-selected inferential member enters that grid. The naive comparator is the project's log-linear pool eq. 6. Under the shared-support, posterior-log-potential, and fixed-weight assumptions of sec. 5.8, it specializes the message-combination term of Friston et al.'s Eq. 7 (2024), rather than reconstructing the complete source protocol. We test it with the matched-pairs Wilcoxon signed-rank test (Wilcoxon 1945) (`statistics.paired_test`, ISC-28), interpreted under the usual signed-rank conditions: seed-level paired differences are independent across the declared seed schedule, while the primary sweep's trial-level diagnostic remains conditional on its fixed world; the differences are ranked after zero differences are removed, and the null is a symmetric distribution of paired differences around zero rather than an equality-of-means claim (Fay and Proschan 2010). This is the right comparison for bounded, often non-Gaussian accuracy deltas, but it is not assumption-free. The test reports the primary matched-pairs rank-biserial effect $r_{\mathrm{rb}} = (T^+ - T^-)/(T^+ + T^-)$. We retain the monotone secondary transform $d_{\mathrm{eq}} = 2r_{\mathrm{rb}}/\sqrt{1 - r_{\mathrm{rb}}^2}$, labeled a *rank-biserial-derived d-equivalent*, never raw Cohen's d and never a replacement for the mean contrast. When r_{rb} saturates at ± 1, the transform diverges; tables print a signed saturation marker rather than a finite million-scale value. The primary-sweep headline display method RKL has rank-biserial effect 1.0000, d-equivalent saturated (r=+1) (large), with the paired mean difference and bootstrap interval reported alongside it.

Every inferential mean — colony free energy, learning-curve KL, per-method accuracy, and the robust-minus-naive accuracy difference — carries a 95% percentile-bootstrap confidence interval (Efron and Tibshirani 1993) from `statistics.bootstrap_ci`, resampled from the declared unit. For the review grid, the unit is the seed after the nested trials have been reduced; for the primary fixed-world sweep, the interval is explicitly trial-level and conditional; for single-seed diagnostics, it is descriptive or trial-level. The single-colony mechanistic rate table is $n = 1$ per cell and is descriptive, not an inferential mean; its companion profile uses the declared nested design. This separation follows simulation-reporting guidance to declare the estimand and the independent Monte Carlo unit (Morris et al. 2019). The interval quantifies variation over the declared resampling unit (seed, recorded step, or matched trial) and is conditional on this simulation design, not on unmodeled real-world deployment uncertainty or alternative contamination models. The headline mean robust-minus-naive accuracy difference is 0.0846 with 95% CI $[0.0821, 0.0873]$, and the accuracy at the verdict rate is 0.9021 for the naive pool (95% CI $[0.8993, 0.9049]$) against 0.9867 for the most accurate robust member (95% CI $[0.9865, 0.9869]$).

For every multi-seed summary we also report the Monte Carlo standard error $(\mathrm{MCSE} = s/\sqrt{n_{\mathrm{seed}}})$ and an approximate two-sided minimum detectable effect (MDE) at the configured target power. These are precision diagnostics over independent seeds, not claims about sampling a real population. The MDE uses a normal approximation conditional on the observed seed-level standard deviation (Koehler et al. 2009); it is reported alongside the non-parametric interval rather than used to replace the paired test.

The sweep compares each robust divergence against the naive pool, so an uncorrected $p < 0.05$ across the family would manufacture false discoveries. We control the false-discovery rate with the Benjamini-Hochberg step-up procedure (Benjamini and Hochberg 1995) (`statistics.bh_fdr`, ISC-29) at $\alpha = 0.05$. The verdict family owns one robust-versus-naive contrast per robust method at the predeclared verdict rate; each rate table owns its own within-method rate family. Families are not pooled across figures or across the review-grid cells. The review-grid families retain all configured method contrasts and do not derive a selected member by pooled mean. The procedure returns both the rejection mask and the monotone BH q-values. BH controls expected false discovery proportion within the stated family, not the family-wise probability of any false positive; the manuscript therefore states the family each table uses. The positive-contrast rule is strict and conjunctive: a method is a BH-rejected positive contrast *iff* BH rejects its null **and** its rank-biserial effect is positive. This is a statistical decision for the named family, not a unique scientific winner. The primary-sweep headline display method has raw $p = 1.11 \times 10^{-158}$, deflating to BH $q = 1.11 \times 10^{-158}$.

Prospective power analysis for the verdict rate I

We report not only *that* the verdict rejected the null but the observed-effect planning power implied by the run and the number of pairs a confirmatory run could budget. The observed-effect design power of the headline paired Wilcoxon at the primary sweep's matched-trial unit, $n_{\mathrm{trial}} = 960$, $\alpha = 0.05$, alternative `greater`, is 1.0000; the prospective sample size for the target power 0.80 at the observed effect is $n_{\mathrm{trial}} = 5$. The estimator is the deterministic noncentral-normal approximation to the matched-pairs t-test, deflated by the Wilcoxon's Pitman asymptotic relative efficiency $3/\pi$, so the reported power is approximately calibrated for the signed-rank test the harness runs (the deflation is exact under a normal-shift alternative); note the power computation is directional for planning, while the reported p-values are two-sided.

Prospective power analysis for the verdict rate I

This is an observed-effect planning approximation conditional on the observed effect size, not independent evidence for the result and not a confirmatory power guarantee.

The results sections render five statistics tables straight from these tokens: the per-rate accuracy sweep (tbl. 3), the robust-versus-naive verdict (tbl. 5), the standardized-effect verdict with observed-effect design power and prospective n (tbl. 6), the per-method accuracy with bootstrap CIs at the verdict rate (tbl. 7), and the per-contamination-rate paired tests (tbl. 4). Every cell is a generated token, never hand-typed.

The honesty contract binds at exactly this tier. The effect-size, CI, and power enrichment above decorates the *server-side* `robust_aggregate` divergence-reweighting contrasts only. It does not certify the per-client β/rcce generalized-Bayes (FedGVI) guarantees, which are pinned by the locked core (Proposition 4 of sec. 6) rather than by these aggregation-level statistics — the three robustness axes of sec. 3.3 kept distinct.

The release now reports computational complexity from the implementation itself. Let N be the number of agents, S the number of categorical states, I the solver-iteration budget, B the number of variational starts, and M the number of conditionally independent observation modalities. The dominant dense work and the storage actually retained by the current NumPy paths are:

Computational complexity and scaling diagnostic I

operation	dominant time order	retained or peak storage
log-linear pooling	$\Theta(NS)$	$\Theta(NS)$
iterative robust pooling	$\Theta(INS)$	$\Theta(NS + IS)$
objective-backed variational pooling	$\Theta(BINS)$	$\Theta(NS + IS + BS)$
self-excluding naive sharing	$\Theta(N^2S)$	$\Theta(NS)$
self-excluding robust sharing	$\Theta(IN^2S)$	$\Theta(NS + IS)$
one-step state inference	$\Theta(MS)$	$\Theta(S)$
server round, excluding queue/network wait	$\Theta(NlogN + INS)$	$\Theta(NS)$

These are dominant interaction counts, not hardware-independent FLOP totals. The N^2 sharing term is material: with sensory attenuation enabled, one round computes one global pool and one leave-one-out pool per agent. The iterative server rules additionally retain their returned per-iteration histories, which is why their storage rows include IS. The server row includes worker-ID sorting and dense aggregation; incoming serialized belief volume is linear in NS, while queue and network latency are deliberately outside this local-compute accounting. The local path serializes one consensus-plus-weight result; if physical broadcast bytes are counted per recipient, that outgoing volume additionally has an N^2 term because each result carries the N agent weights.

The accompanying seeded benchmark measures the real public call paths on the declared grids $N \in \{4, 8, 16, 32, 64\}$, $S \in \{256, 512, 1024, 2048, 4096\}$, self-excluding sharing $N \in \{4, 8, 16, 32\}$, and $M \in \{1, 2, 4, 8\}$. The fixed dimensions are $N = 256$, $S = 64$ for aggregation, and $S = 16384$ for state inference; direct aggregation and server timings use $I = 6$, while the public robust self-excluding sharing path uses its solver budget $I = 32$; the default variational path uses $B = 3$. Each grid point is warmed up 1 time(s) and measured 5 time(s), with median time plotted and the observed repeat range shown as a min–max bar. The fixed input seed is 20260728 on the $arm64$ machine using Python $3.13.11$ and NumPy $2.4.2$.

The measured log–log slopes are descriptive checks of the expected orders, not performance guarantees: agent-axis slopes are 0.89 (log-linear), 0.94 (iterative robust), 0.71 (variational), 1.59 (naive self-excluding sharing), and 1.93 (robust self-excluding sharing); state-axis slopes are 0.42, 0.40, and 0.41; the modality-axis inference slope is 0.67. The slope fit is a timing diagnostic on this machine, not an inferential test and not evidence that the same constants hold under another BLAS, accelerator, process topology, or distributed network. A finite grid can also yield a sublinear fitted slope when validation, allocation, cache, and interpreter overheads are material; the implementation-derived order is the governing claim, not equality between a finite-grid slope and its exponent.

Figure fig. 1 visualizes the implementation-derived orders and the corresponding finite-grid timing diagnostic.

Computational complexity and scaling diagnostic I

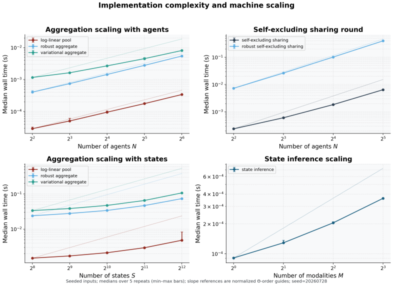

Figure 1: Implementation-derived complexity and seeded machine-scaling diagnostic. Source relation: original project computational-complexity diagnostic; estimand: median wall-clock time of the real categorical aggregation, naive and robust self-excluding sharing, and state-inference call paths as the declared dimension changes; uncertainty: min–max span over the repeated timings, not a confidence interval; replication unit: fixed seeded input at each grid point with the declared timing repeats. The x-axis is the varied agent, state, or modality dimension, and the y-axis is median wall-clock time in seconds. The panels show agent scaling for the aggregation rules, naive and iterative-robust N^2 leave-one-out sharing, state scaling for the aggregators, and modality scaling for one-step inference. Dotted lines are normalized Θ-order guides from the implementation-derived accounting; they are not fitted claims. The experiment ran on $arm64$ with Python $3.13.11$, NumPy $2.4.2$, seed 20260728, and 5 measured repeat(s) after 1 warmup(s).